

PROJECT DOCUMENTATION

Kenya Product Sales

KPI Dashboard

An interactive Tableau Public dashboard tracking revenue, profit, units sold, and COGS across Kenyan counties (2020-2022).

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PLATFORM	Tableau Public
PERIOD	Q1 2020 – Q3 2022
GEOGRAPHY	Kenya (County-Level)
REPOSITORY	github.com/skynet-datagrid-labs/Kenya-Product-Sales

Kenya Product Sales – KPI Dashboard

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1. Project Overview

This project is a business intelligence solution developed for an FMCG company operating across the Kenyan market. The dashboard was built using Tableau Public and is designed to give business stakeholders a single, unified view of sales performance across all counties, product categories, and customer segments for the period Q1 2020 through Q3 2022.

The central goal of the project is to move beyond static reports and enable interactive, data-driven decision-making. A stakeholder can open the dashboard, select a specific quarter range, filter by product category or company, and immediately see how revenue, profit, units sold, and cost of goods sold respond across all visual components simultaneously. This kind of cross-filtered, dynamic exploration is what makes the tool genuinely useful in a business environment.

2. Business Context

Kenya's FMCG sector is highly competitive, with major international brands such as Coca-Cola and Dr. Pepper operating alongside local producers like Crown Beverages and Kericho Gold across beverage categories including soft drinks, coffee, tea, and alcoholic beverages. Understanding which products are generating revenue, which counties are underperforming, and which customers contribute most to the bottom line is critical for sales strategy, distribution planning, and marketing allocation.

This dashboard directly addresses that need by consolidating county-level geographic data with transactional sales records into a single interactive interface, built on a shapefile of Kenyan county boundaries and a structured sales CSV dataset.

3. Key Performance Indicators

The following aggregate metrics cover the full period Q1 2020 to Q3 2022:

Metric	Aggregate Value (Full Period)
Total Revenue	47,712
Total Profit	21,708
Total Units Sold	4,921
Cost of Goods Sold	26,004
Profit Margin (approx.)	45.5%

4. Dashboard Components

The dashboard is composed of six interconnected visual components, each of which responds dynamically to the filter selections made in the control panel:

- **Revenue by Month:**A line chart tracking monthly revenue from Q1 2020 through Q3 2022, allowing stakeholders to identify seasonal trends and growth periods.
- **Revenue by County:**A choropleth map of Kenya rendered using the county-level shapefile, showing revenue concentration geographically and making regional disparities immediately visible.
- **Revenue by Product:**A horizontal bar chart ranking top-selling products such as Stoney Tangawizi, Dormans Coffee, and Kericho Gold by total revenue contribution.
- **Revenue by Customer:**A bar chart ranking individual customers by revenue contribution, useful for identifying key accounts and high-value relationships.
- **Dynamic Metric Toggle:**A parameter control allowing the user to switch the entire dashboard view between Revenue, Profit, Units Sold, and Cost of Goods Sold in a single click.
- **Multi-Filter Panel:**A set of filters enabling drill-down by quarter range, company, and product category (Alcoholic, Coffee, Soft Drinks, Tea), with all views updating simultaneously.

5. Data Sources

The project is built on two primary data files stored in the data/ directory of the repository:

File	Type	Description
kenya_counties_accurate.shp	Shapefile	County-level geographic boundaries for Kenya, used to render the choropleth map in Tableau.
soft_drink_sales_kenya.csv	CSV	Transactional sales records containing revenue, profit, units sold, COGS, product, company, customer, county, and date fields.

6. Data Scope

Dimension	Details
Time Period	Q1 2020 through Q3 2022
Geography	Kenya, broken down to county level
Categories	Alcoholic Beverages, Coffee, Soft Drinks, Tea
Companies	Coca-Cola, Crown Beverages, Dr. Pepper, Kericho Gold, and others
Metrics	Revenue, Profit, Units Sold, Cost of Goods Sold

7. Repository Structure

The repository is organised as follows:

```
Kenya-Product-Sales/
|
+-- data/
|   +-- kenya_counties_accurate.shp    # County boundary shapefile
|   +-- soft_drink_sales_kenya.csv     # Sales transaction records
|
+-- assets/
|   +-- SalesKPIKenya.gif              # Dashboard preview animation
|
+-- workbook/
|   +-- KenyaSaleskpi.twb            # Tableau workbook file
|
+-- README.md                         # Project overview
```

8. How to Use

1. Clone the repository from GitHub:
github.com/skynet-datagrid-labs/Kenya-Product-Sales
2. Open the Tableau workbook file (KenyaSaleskpi.twb) in Tableau Public Desktop.
3. If prompted, refresh the data source and point it to the CSV and shapefile in the data/ folder.
4. Use the quarter range slicer to select a time window of interest.
5. Use the company and category filters to narrow down the view.
6. Toggle between Revenue, Profit, Units Sold, and COGS using the metric selector.
7. Hover over any county on the map or bar in a chart to see detailed tooltips.

9. Technical Requirements

Requirement	Details
Tableau Public Desktop	Free. Available at https://public.tableau.com/app/discover
Operating System	Windows or macOS (Tableau Public Desktop requirement)
Data Files	Both files in data/ must remain in the same relative path as the workbook

10. Contributing & License

This project is open source and contributions are welcome. If you would like to report an issue, suggest an improvement, or submit a pull request, please do so through the GitHub repository. The project is licensed under an open source licence – refer to the LICENSE file in the repository root for the full terms.

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